

ProcessBiologyStep temp

1. Thermal Performance
SimSpecies.CalculatePerformance
Arrhenius + CTmin/CTmax
cosine fade
RawThermalPerf; ThermalPerf =
Raw × Pmax

2. Feeding / Predation
Tier 1: FedRate = min(1,
HE × food_density), linear
(v10)
Tier 2: Holling II + per-
predator FedRate (v11)
fedRate_i = min(1,
huntingSuccess_i ×
scarcityFactor)
_predationAccumulators
on prey

3. Raw Final Performance
RawFinalPerf = Raw ×
FedRate
(Condition drain target –
varies with food density
for Tier 1 in v10)

4. Update Condition
asymmetric drift toward
target
rates scaled by Pmax
(pmaxSafe)

5. Final Performance
FinalPerf = ThermalPerf ×
FedRate
(computed for logging; not
read downstream)

6. Thermal Death
instant wipeout when
RawThermalPerf == 0

7. Condition Death
graduated severity below
DeathThreshold
survivor fitness boost
_conditionDeathAccumulators

8. Reproduction (v10: no
soft cap on births)
reproScale(Condition) ×
Pmax
Tier 1
NO_PREDATOR_PENALTY
only
_birthAccumulators
newborns inherit parent
group Condition

9. Natural Death
flat NaturalDeathRate ±
variance
_naturalDeathAccumulators

10. Population Rounding
round away-from-zero

End step